WP1 — FCLC Federated Clinical Learning Cooperative

Scope for Fraunhofer IESE Technical Partnership

EIC Pathfinder Open 2026 · CommonHealth · €0.6M · 12 months · TRL 3→5

Objective

Build a **production-grade, EU-compliant federated learning infrastructure** for multi-institutional clinical data analysis, achieving system-level privacy guarantee ε ≤ 1.0 and demonstrating functionality in a real-world clinical pilot (WP5 — anemia cohort N=50) by Month 24.

Technical stack (current v6.2, to be hardened)

Layer	Technology	Current status	Target
Secure aggregation	SecAgg+ with ChaCha20-Poly1305 + SHA-256 seed + Shamir GF(257) dropout recovery	Implemented, 96/96 tests pass	Production audit
Differential privacy	Laplace mechanism + Rényi DP accounting	ε = 2.0 (research prototype)	**ε ≤ 1.0** (compliance target)
Coordination layer	Elixir/Phoenix LiveView	Functional	Multi-node scaling
Data schema	BioSenseExport JSON, FHIR-compatible	Specified	FHIR-Starter integration
Compliance	GDPR Art. 28 DPA template	Drafted	EU AI Act Art. 22 alignment

Tasks

Task	Months	Lead	Effort
T1.1 Architecture refinement + security threat model	1-6	Fraunhofer IESE (proposed)	6 PM
T1.2 DP hardening	3-12	Fraunhofer + GLA	10 PM

Task	Months	Lead	Effort
to ε ≤ 1.0 (PATE/DP-SGD redesign)			
T1.3 Multi-node production deployment (3-node Synthea demo)	6-15	GLA + Fraunhofer	8 PM
T1.4 WP5 clinical integration (anemia cohort FL)	12-24	GLA clinical + Fraunhofer	6 PM

Deliverables

- **D1.1** (M6): Architecture specification + threat model
- **D1.2** (M12): FCLC v2.0 open-source release (GitHub)
- **D1.3** (M18): Independent cryptographic + privacy audit report
- **D1.4** (M24): Multi-institutional clinical deployment demo

Milestones

- **M1.1** (M6): Target ε ≤ 1.0 achieved on synthetic data
- **M1.2** (M18): 3-node production demo complete
- **M1.3** (M24): WP5 clinical integration live (Aqtivirebuli anemia pilot)

Role for Fraunhofer IESE

Co-lead WP1 with Georgia Longevity Alliance. Primary responsibilities:

1. Production-grade software engineering rigor (CI/CD, compliance testing)
2. Regulatory alignment (EU AI Act Art. 22 for high-risk medical AI)
3. FHIR/medical data standards integration (leverage FHIR-Starter expertise)
4. Cryptographic audit coordination with partner (KU Leuven COSIC proposed)
5. Technical deliverables D1.1-D1.4

Budget allocation (indicative)

Category	%
Personnel (software engineers + researchers)	65%
Equipment (compute, test infrastructure)	5%

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Travel (consortium meetings, conferences)	10%
Subcontracting (external audit)	15%
Other (dissemination, overhead)	5%

Risks

Risk	Impact	Mitigation
$\varepsilon \leq 1.0$ target technically challenging	High	PATE/DP-SGD hybrid redesign in T1.2
EU AI Act Art. 22 evolving interpretation	Medium	Fraunhofer regulatory expertise; quarterly review
Clinical data availability (WP5 dependency)	Medium	Synthea synthetic data for pre-deployment testing

Why Fraunhofer IESE

- Institute mission alignment: Digital Health + medical AI software engineering
- Existing project reference: FHIR-Starter (AI-based software for structuring medical data)
- Production-grade engineering rigor — critical differentiator vs. academic-only partners
- Regulatory compliance track record in EU medical AI context
- Geographic/institutional independence from Host (strengthens Consortium quality section)

This scope is non-binding at LoI stage. Detailed PM allocation, budget, and deliverable refinement will follow Grant Agreement negotiation.